EXPERIMENT 4: IMPLEMENT PACKET SNIFFING USING RAW SOCKET IN PYTHON

AIM:

To capture and inspect packet headers using a raw socket in Python for educational network analysis.

PROCEDURE:

1. Install Python on a Linux system.

2. Save the program below as packet_sniffer.py.

3. Run it with administrator/root privileges because raw sockets require elevated permissions.

4. Generate permitted test traffic on the local network and observe packet header information.

5. Stop the program with Ctrl+C. Capture only traffic on networks where permission has been granted.

6. Confirm that the operating system and network interface support the selected raw-socket operation.
7. Run the program only on a system and network for which permission has been granted.
8. Compare source address, destination address, protocol number, packet length and TTL across captured packets.

CODE:

```
import socket
import struct

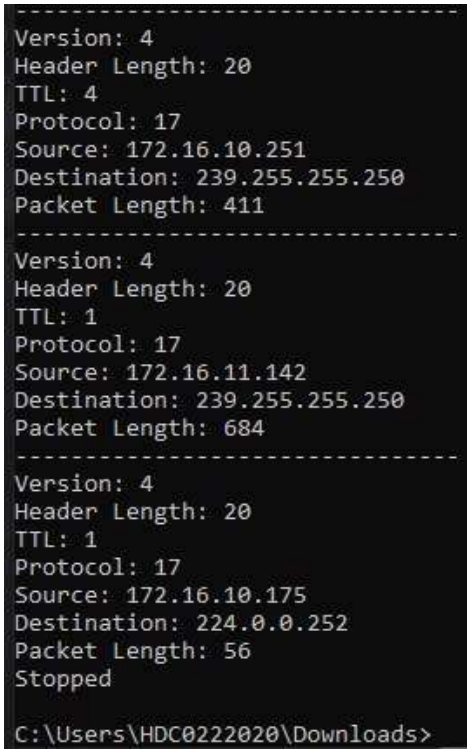
def ipv4_header(packet):
    if len(packet) < 20:
        return
```

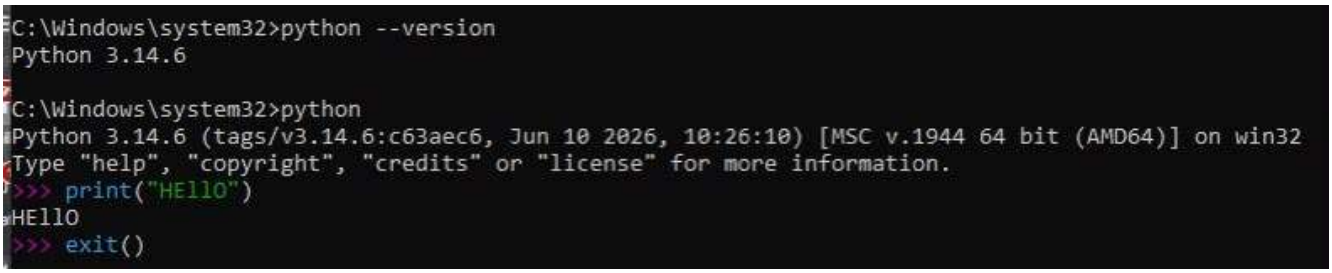
```
    version_ihl = packet[0]
    version = version_ihl >> 4
    ihl = (version_ihl & 0x0F) * 4
    if version != 4 or len(packet) < ihl:
        return
    total_length = struct.unpack("!H", packet[2:4])[0]
    ttl = packet[8]
    protocol = packet[9]
    src = socket.inet_ntoa(packet[12:16])
    dst = socket.inet_ntoa(packet[16:20])
    print(f"IPv{version} {src} -> {dst},
protocol={protocol}, length={total_length}, TTL={ttl}")

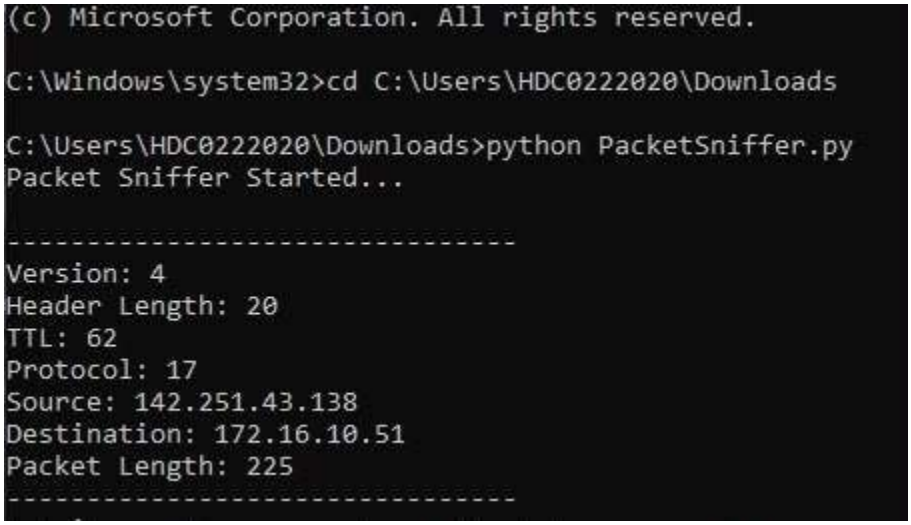
sock = socket.socket(socket.AF_INET, socket.SOCK_RAW,
socket.IPPROTO_IP)
sock.bind(("0.0.0.0", 0))
sock.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)

print("Listening for IPv4 packets. Press Ctrl+C to
stop.")
try:
    while True:
        packet, address = sock.recvfrom(65535)
        ipv4_header(packet)
except KeyboardInterrupt:
    print("\nCapture stopped.")
finally:
    sock.close()
```

OUTPUT:


Version: 4
Header Length: 20
TTL: 4
Protocol: 17
Source: 172.16.10.251
Destination: 239.255.255.250
Packet Length: 411
Version: 4
Header Length: 20
TTL: 1
Protocol: 17
Source: 172.16.11.142
Destination: 239.255.255.250
Packet Length: 684
Version: 4
Header Length: 20
TTL: 1
Protocol: 17
Source: 172.16.10.175
Destination: 224.0.0.252
Packet Length: 56
Stopped
C:\Users\HDC0222020\Downloads>


C:\Windows\system32>python --version
Python 3.14.6
C:\Windows\system32>python
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("HEllO")
HEllO
>>> exit()


(c) Microsoft Corporation. All rights reserved.
C:\Windows\system32>cd C:\Users\HDC0222020\Downloads
C:\Users\HDC0222020\Downloads>python PacketSniffer.py
Packet Sniffer Started...
Version: 4
Header Length: 20
TTL: 62
Protocol: 17
Source: 142.251.43.138
Destination: 172.16.10.51
Packet Length: 225

RESULT:

Thus, a Python raw-socket program was used to inspect IPv4 packet header fields in a permitted test environment.